A Brief Overview of Quantum Computation and Quantum Information

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Some Notes Before the Talk…

- I know nothing about physics. Luckily, we don’t need to
- You don’t need to know quantum physics to do quantum computing, just like you don’t need semiconductor physics to program a classical computer.
- All you need to know is a few “postulates” of quantum mechanics

Outlines

- Concepts and Postulates of Quantum Mechanics
  - Postulate 1: A quantum bit
  - Postulate 2: Measurement
  - Postulate 3: Evolution of quantum systems
  - Postulate 4: Multi-qubit systems
- Computation
  - Creation of EPR pairs
  - "Spooky Action at a Distance"

Understanding 1 Quantum Bit (Qubit)

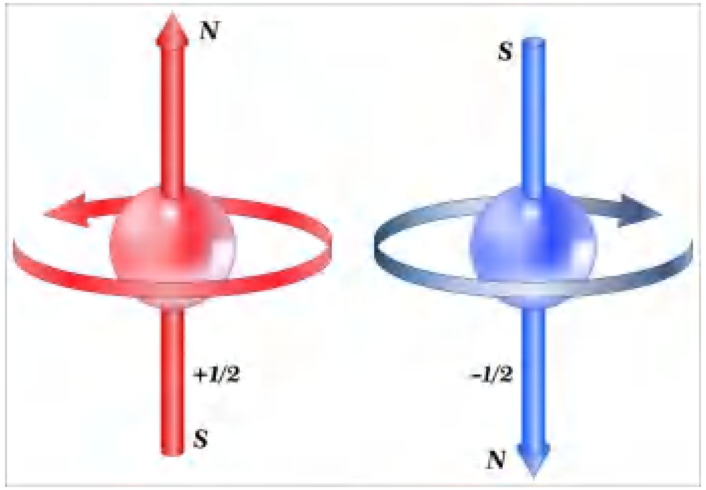
- In Classical Computing:

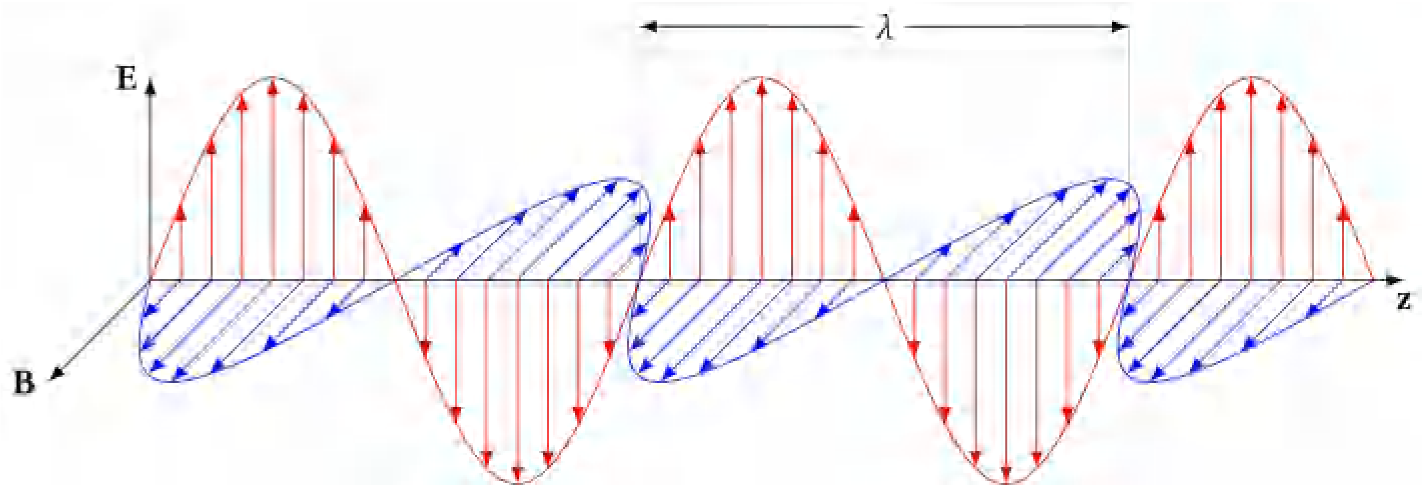
Logical bit:	Physical implementation:
0	Low voltage
1	High voltage

- Using millions of electrons store 1 bit
- Question: Can we make a bit out of 1 particle, e.g., an electron or photon?
- Understanding properties of such particles is the domain of quantum mechanics

Understanding 1 Quantum Bit (Qubit)

- We need a particle with 2 possible “properties”, which we can call 0 or 1





Electron “**spin**”:
Up: 0
Down: 1

Photon “**polarization**”:
Horizontal: 0
Vertical: 1

Postulate 1: A Quantum Bit (Qubit)

- If a quantum system/particle can be in of two **basic states** $|0\rangle$ or $|1\rangle$, it can also be in a **superposition** state:

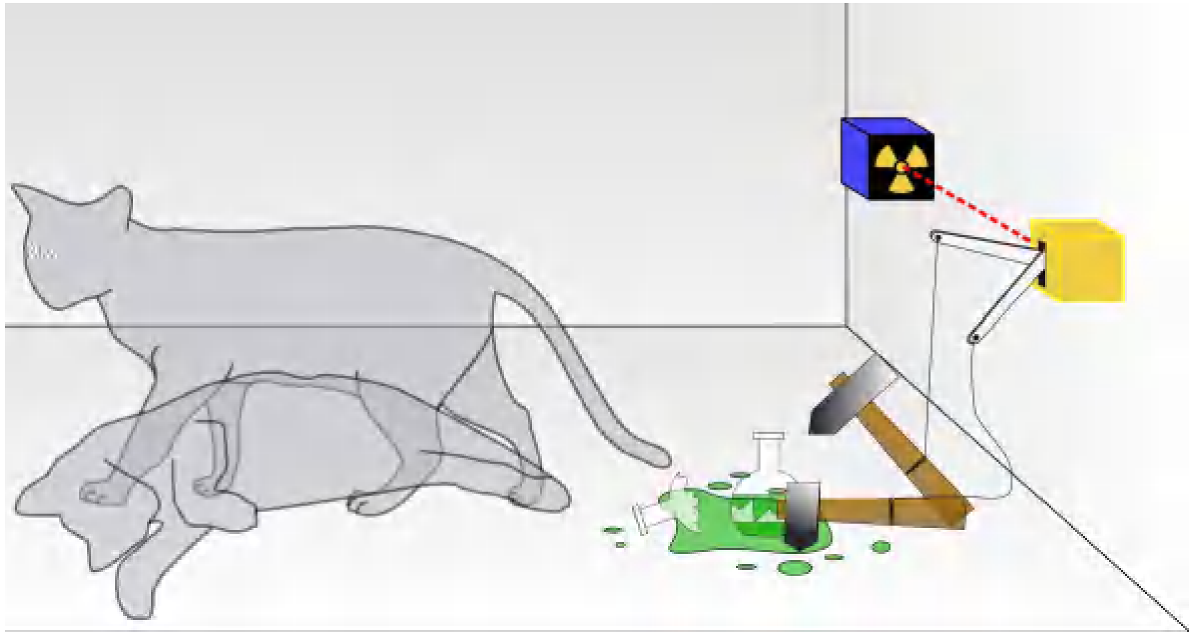
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

I.e., α "**amplitude**" on $|0\rangle$, β "**amplitude**" on $|1\rangle$, where α and β are numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$

- $|\psi\rangle$, $|0\rangle$ and $|1\rangle$: Dirac's Bra-Ket notation, explain later
- Superposition: a particle can exist in multiple states at the same time
- Amplitude: just a number, can be negative

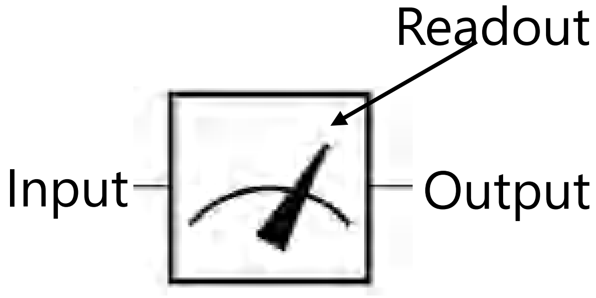
Examples of Superposition States

- Example 1: a photon may have state $0.8|0\rangle - 0.6|1\rangle$
  - 0.8 amplitude on $|0\rangle$
  - -0.6 amplitude on $|1\rangle$
  - Check: $0.8^2 + (-0.6)^2 = 0.64 + 0.36 = 1$
- Example 2: Schrödinger's cat



Postulate 2: Measurement

- For a particle with amplitude α on $|0\rangle$, β on $|1\rangle$, i.e., $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, if you measure it, then
  - With probability $|\alpha|^2$, readout shows $|0\rangle$
  - With probability $|\beta|^2$, readout shows $|1\rangle$



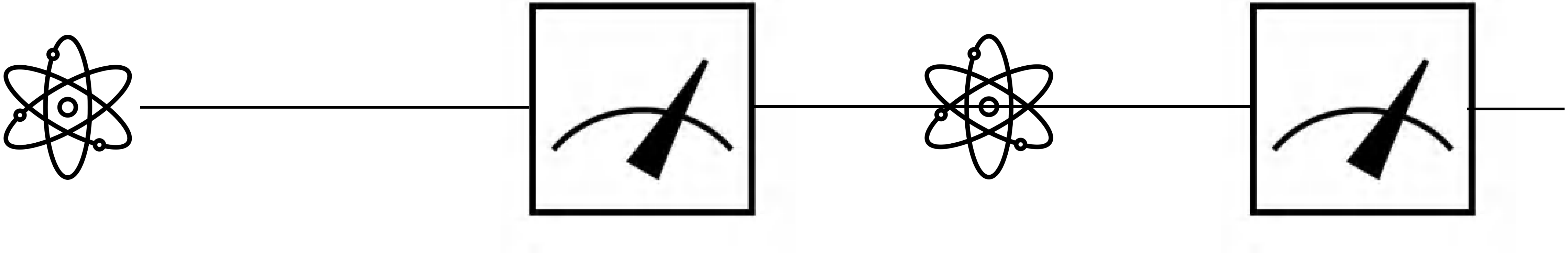
- And,
  - If readout is $|0\rangle$, particle's state "collapses" to "1 amplitude on $|0\rangle$, 0 amplitude on $|1\rangle$"
  - If readout is $|1\rangle$, particle's state "collapses" to "0 amplitude on $|0\rangle$, 1 amplitude on $|1\rangle$"
- Measurements can change quantum states!

Postulate 2: Measurement

- Example:

Readout:
64% chance of $|0\rangle$,
36% chance of $|1\rangle$

Readout:
100% chance of $|0\rangle$,
0% chance of $|1\rangle$

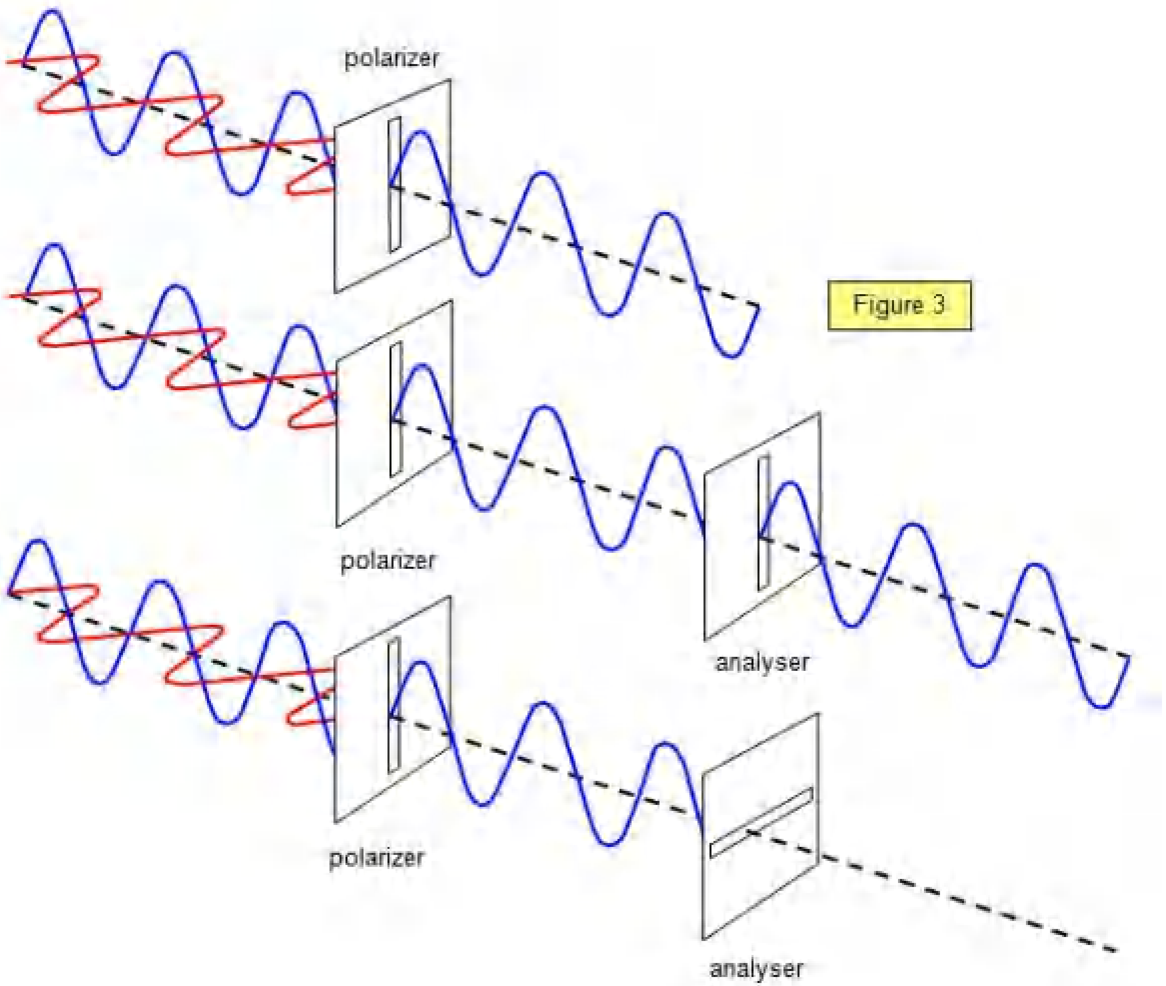


Init state:
0.8 on $|0\rangle$, -0.6 on $|1\rangle$

State becomes
(for example):
1 on $|0\rangle$, 0 on $|1\rangle$

Postulate 2: Measurement

- This is best illustrated using polarizing filters (e.g., 3d glasses)



General Forms of Postulates 1 and 2

- It is perfectly possible that a particle has more than 2 basic states: $|1\rangle$, $|2\rangle$, $|3\rangle$, $|4\rangle$, …
- Postulate 1: a general state is
  - Amplitude α_1 on $|1\rangle$
  - Amplitude α_2 on $|2\rangle$
  - Amplitude α_3 on $|3\rangle$
  - ……
  - Amplitude α_d on $|d\rangle$
- Such that $|\alpha_1|^2 + \ldots + |\alpha_d|^2 = 1$
- Postulate 2: after measurement, readout $|i\rangle$ with probability $|\alpha_i|^2$, and the state collapses to 1 amplitude on $|i\rangle$, 0 amplitude on the rest

Called "qu**d**it"

Time to Begin the Math Properly!

- For a qudit, we need to track d numbers whose squared magnitudes sum to 1. This is nothing more than a d-dimensional unit vector:

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{\mathrm{d}} \end{pmatrix} \in \mathbb{C}^d, \text{ with } |\alpha_1|^2 + \ldots + |\alpha_d|^2 = 1$$

- Any isolated physical system is completely described by a complex unit vector

Dirac's Bra-Ket Notation

- In linear algebra, we have learned:

$$\begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} = 3e_1 + 5e_2 - 2e_3, \text{ where}$$

$$|1\rangle = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |2\rangle = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |3\rangle = e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- Exception when $\mathrm{d} = 2$, $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- Qudit state: $\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{\mathrm{d}} \end{pmatrix} = \alpha_1|1\rangle + \alpha_2|2\rangle + \ \ldots\ + \alpha_{\mathrm{d}}|\mathrm{d}\rangle$, , with $|\alpha_1|^2 + \ldots + |\alpha_d|^2 = 1$

Dirac's Bra-Ket Notation

- Notation: $|\text{any name}\rangle$, or $\langle\text{any name}|$
- $|\cdot\rangle$ signifies a **column vector**, called a "ket"
- $\langle\cdot|$ denotes its conjugate transpose, i.e., $|\cdot\rangle^{\dagger}$, a row vector, called a "bra"
- Example:
- $|\phi\rangle = \begin{pmatrix} 3 \\ 5+\mathrm{i} \\ -2 \end{pmatrix}, \langle\phi| = (3,5-i,-2)$

Postulate 3: Evolution of Quantum Systems

- A qudit's state can be changed by any **linear transformation** that **preserves lengths**
  - Linear transformation: matrices
  - Preserving lengths: obviously necessary due to Postulate 1

- Which are the linear transformation that preserves all lengths? They are rotations/reflections
- In linear algebra, we call them "Unitary Transformations"
- Definition: U is unitary $\Leftrightarrow U^{\dagger}U = I \Leftrightarrow UU^{\dagger} = I \Leftrightarrow U^{-1} = U^{\dagger}$

Postulate 3: Evolution of Quantum Systems

- Examples of unitary matrices/gates/operations:

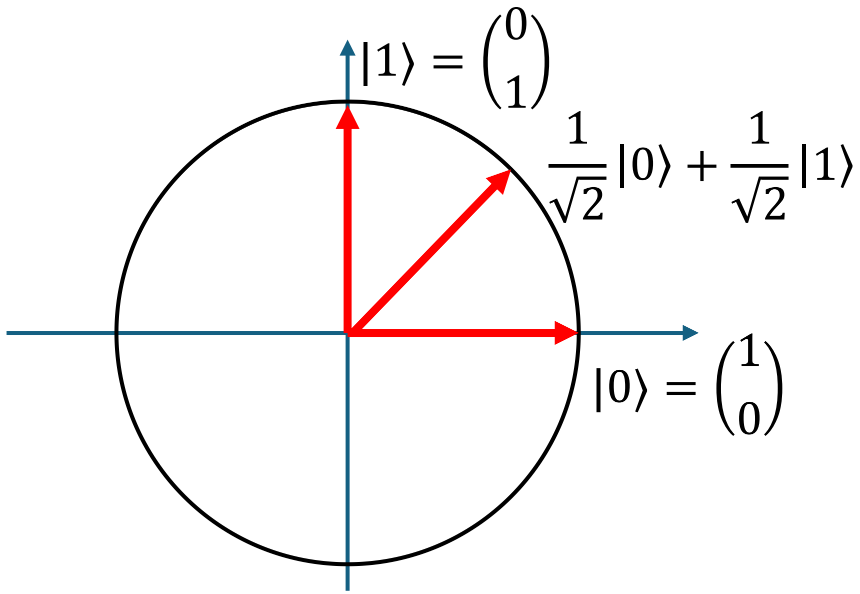
$$\text{Hadamard: } H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\text{NOT: } \mathrm{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- Example of applying a NOT gate
- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- $|\psi'\rangle = \mathrm{X}|\psi\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix} = \beta|0\rangle + \alpha|1\rangle$

Postulate 3: Evolution of Quantum Systems

- Example of applying a Hadamard gate
- $|\psi\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- $|\psi'\rangle = H|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$



Postulate 4: Multi-Qubit Systems

- We have two photons:

$$\alpha_0|0\rangle + \alpha_1|1\rangle$$
$$\beta_0|0\rangle + \beta_1|1\rangle$$

- What is the joint state of the 2-qubit system?
- What you guess is correct. It is of the form:

$$\alpha_0\beta_0|00\rangle + \alpha_0\beta_1|01\rangle + \alpha_1\beta_0|10\rangle + \alpha_1\beta_1|11\rangle$$

- Check:
  $|\alpha_0\beta_0|^2 + |\alpha_0\beta_1|^2 + |\alpha_1\beta_0|^2 + |\alpha_1\beta_1|^2 = (|\alpha_0|^2 + |\alpha_1|^2)(|\beta_0|^2 + |\beta_1|^2) = 1 \times 1 = 1$

Postulate 4: Multi-Qubit Systems

- More generally:

The joint state is $d \times e$ dimensional!

Alice's d-dimensional system

Bob's e-dimensional system

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_\mathrm{d} \end{pmatrix} \otimes \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_\mathrm{e} \end{pmatrix} \Longrightarrow \begin{pmatrix} \alpha_1\beta_1 \\ \alpha_1\beta_2 \\ \vdots \\ \alpha_1\beta_\mathrm{e} \\ \alpha_2\beta_1 \\ \alpha_2\beta_2 \\ \vdots \\ \alpha_\mathrm{d}\beta_\mathrm{e} \end{pmatrix}$$

This is called "Tensor Product"

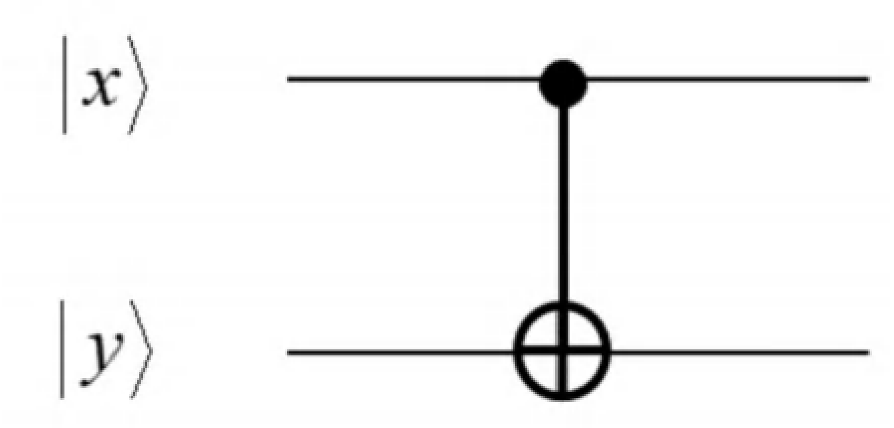
Evolution of Multi-Qubit Systems

- Still unitary transformations, but with higher dimensions.
- Example:

$$\mathrm{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

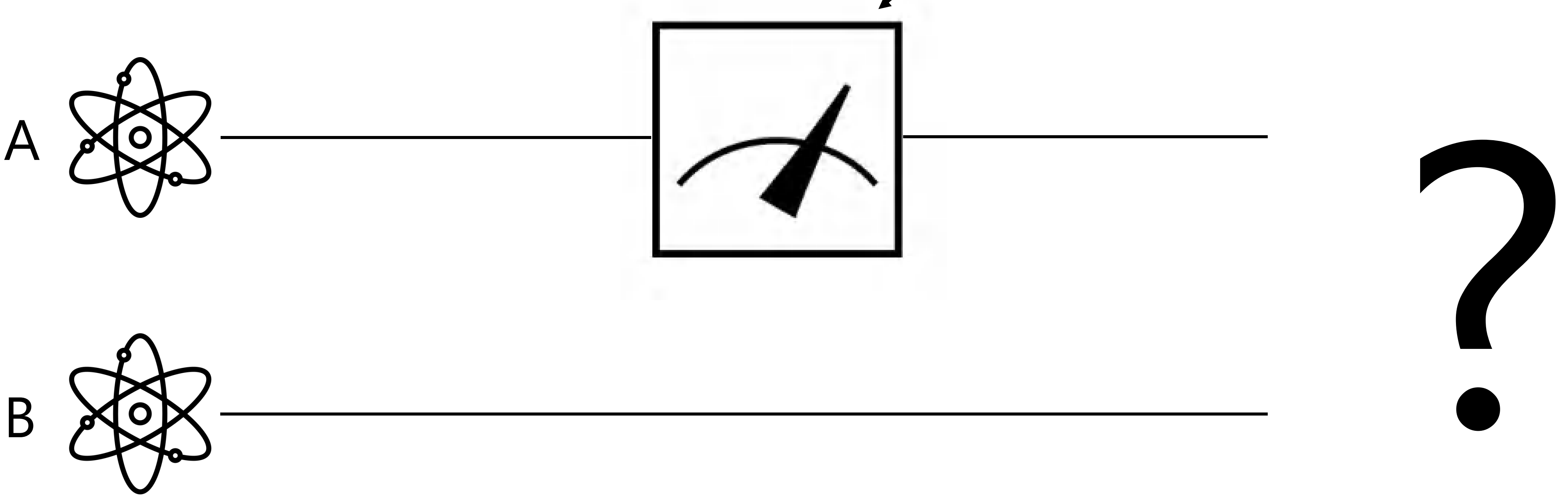
- It is easy to verify:

$$CNOT|00\rangle = |00\rangle$$
$$CNOT|01\rangle = |01\rangle$$
$$CNOT|10\rangle = |11\rangle$$
$$CNOT|11\rangle = |10\rangle$$



Partial Measurements

Readout $|0\rangle$ or $|1\rangle$



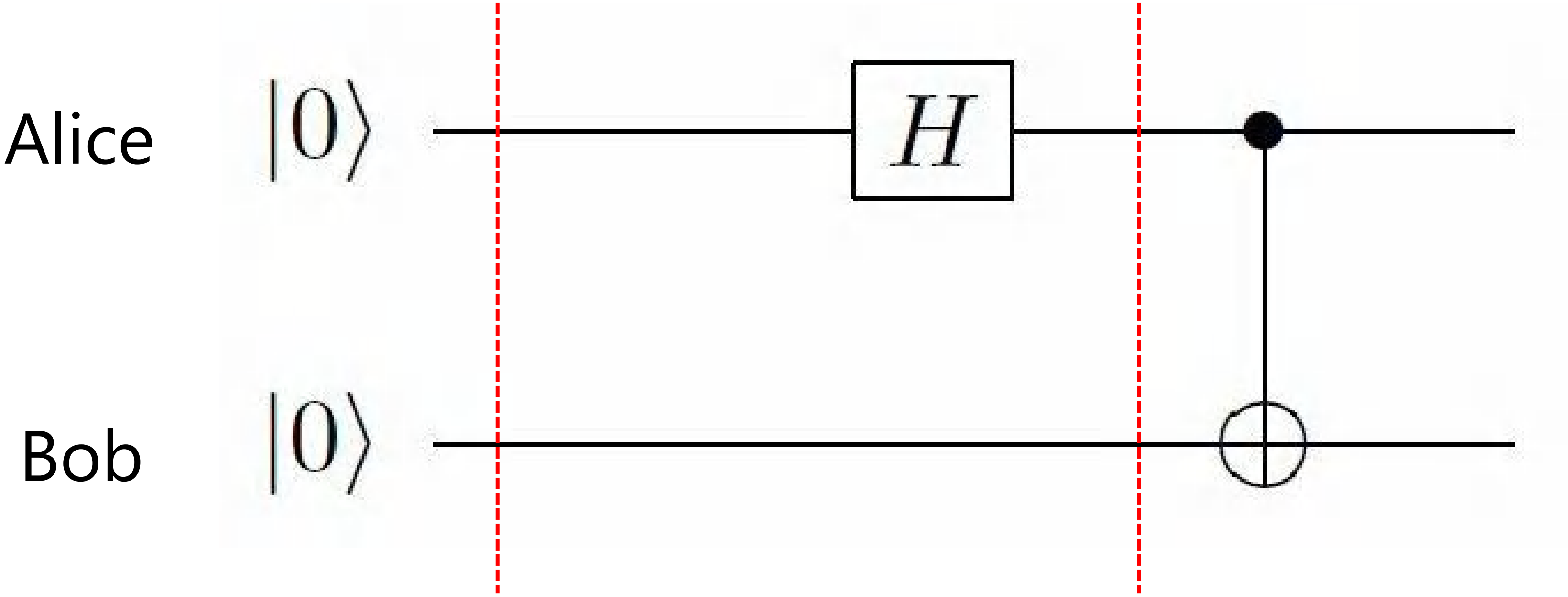
Joint state:
$\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$

Partial Measurements

- If you measure the first qubit of state $\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$
- With probability $P_0 = |\alpha_{00}|^2 + |\alpha_{01}|^2$, readout "$|0\rangle$" , and the joint state "collapses" to $\frac{\alpha_{00}|00\rangle+\alpha_{01}|01\rangle}{\sqrt{P_0}}$
- With probability $P_1 = |\alpha_{10}|^2 + |\alpha_{11}|^2$, readout "$|1\rangle$" , and the joint state "collapses" to $\frac{\alpha_{10}|10\rangle+\alpha_{11}|11\rangle}{\sqrt{P_1}}$

That's All the Math You Need to Know
Let's Do Some Computations…

- Create an EPR (Einstein, Podolsky, Rosen) pair:



$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Called "EPR Pair"

$$|0\rangle \otimes |0\rangle = |00\rangle$$

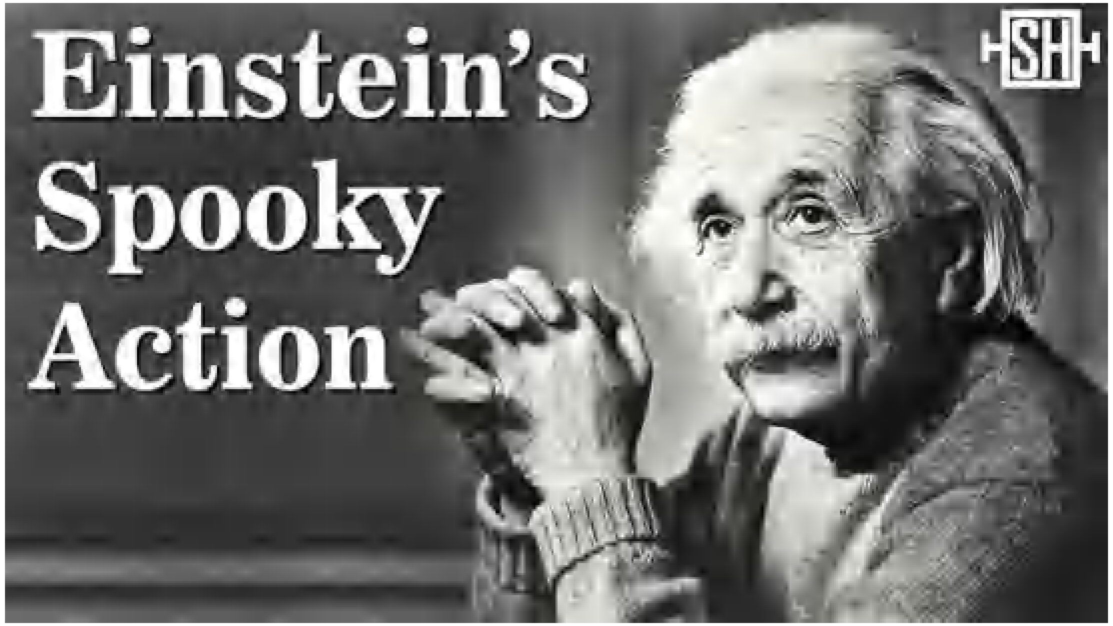
$$\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle$$
$$= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

Something Strange Happens

- Suppose Alice and Bob have an EPR pair $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$
- In principle, they can take their particles far apart
- Alice goes to Mars, and measures her qubit
- E.g., with $\left(\frac{1}{\sqrt{2}}\right)^2 = 0.5$ chance, she reads out "$|0\rangle$", and the joint state "collapses" to $|00\rangle$
- This means Bob's state "instantaneously" becomes $|0\rangle$, and Alice knows that Bob will get "$|0\rangle$" if he measures his qubit
- Conversely, now if Bob measures on Earth, he instantly knows what Alice's readout was

“Spooky Action at a Distance”

- There is a general principle in physics that information should not be communicated faster than the speed of light
- This is “crazy”. Einstein didn’t believe it



Some History

- In 1935, Einstein argued that there could be “hidden variables”
  - Quantum mechanics only gives statistical predictions
  - Each particle should have definite physical properties
  - The apparent randomness is just because we don’t know these hidden parameters
- In 1964, John Bell proposed Bell inequality and showed that:
  - If there are “hidden variables”, Bell inequality holds
  - Otherwise, Bell inequality does not hold
- A series of experiments tested Bell inequality
  - Alain Aspect (1982), Hanson (2015)
  - All experiments showed clear violations of Bell inequalities
- Local hidden variable theories are ruled out
- In 2022, the Nobel Prize in Physics was awarded to Alain Aspect

Thanks for Listening

Q/A